WORKED SOLUTION

STEP III 2017, Question 6

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

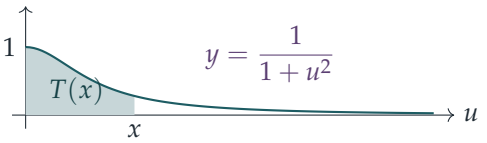
step.maths.org/questions/17-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 4.2, *Building Trigonometry from an Integral*: https://beyondstep.keithmcnulty.org/ch04/trig-from-integral.html

The key idea

We are building the inverse tangent from nothing but its integral, so every fact must come from a substitution. The substitution $u \mapsto 1/u$ swaps the ranges $(0, x)$ and $(1/x, \infty)$, and the Möbius substitution $v = \frac{u+a}{1-au}$ leaves the integrand-with-differential $\frac{\mathrm{d}u}{1+u^2}$ unchanged, so it simply shifts the limits of integration. Choosing the numbers in the final part so that the shifted limit lands back on a value we already know produces an equation we can solve.

Solution



$T(x)$ is the shaded area; T_∞ is the whole area under the curve for $u \geq 0$.

Part (i)

Substitute $u = 1/t$ in the integral for $T(x)$, so $\mathrm{d}u = -t^{-2}\,\mathrm{d}t$; as u runs from 0 to x, t runs from ∞ down to $1/x$. Then

$$\begin{aligned} T(x) &= \int_{\infty}^{1/x} \frac{1}{1+t^{-2}}\left(-\frac{1}{t^2}\right)\mathrm{d}t = \int_{1/x}^{\infty} \frac{1}{t^2+1}\,\mathrm{d}t \\ &= \int_0^{\infty} \frac{\mathrm{d}t}{1+t^2} - \int_0^{1/x} \frac{\mathrm{d}t}{1+t^2} = T_\infty - T(x^{-1}). \end{aligned}$$

(The integral starting at 0 is a proper integral, because T_∞ is finite, so splitting the range is legitimate.) This is the required result.

Part (ii)

The derivative. By the quotient rule, for $u \neq a^{-1}$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{(1-au)+a(u+a)}{(1-au)^2} = \frac{1+a^2}{(1-au)^2}.$$

On the other hand,

$$1+v^2 = \frac{(1-au)^2+(u+a)^2}{(1-au)^2} = \frac{1+a^2u^2+u^2+a^2}{(1-au)^2} = \frac{(1+a^2)(1+u^2)}{(1-au)^2}.$$

Dividing by $1+u^2$ gives $\dfrac{1+v^2}{1+u^2} = \dfrac{1+a^2}{(1-au)^2} = \dfrac{\mathrm{d}v}{\mathrm{d}u}$, as required.

The addition formula. Take $a > 0$ and $0 < x < \frac{1}{a}$. For $0 \le u \le x$ we have $1-au \ge 1-ax > 0$, so v is defined and continuous on the whole range, and $\frac{\mathrm{d}v}{\mathrm{d}u} > 0$, so v increases from $v(0) = a$ to $v(x) = \frac{x+a}{1-ax}$.

? **Ask yourself** Why does it matter that $x < \frac{1}{a}$?

If x reached $\frac{1}{a}$, then v would blow up inside the range of integration and the substitution would be meaningless. With $x < \frac{1}{a}$ it is a smooth, increasing change of variable on $[0, x]$.

The derivative result says exactly that $\dfrac{1}{1+u^2} = \dfrac{1}{1+v^2}\dfrac{\mathrm{d}v}{\mathrm{d}u}$, so

$$\begin{aligned} T(x) = \int_0^x \frac{\mathrm{d}u}{1+u^2} &= \int_a^{\frac{x+a}{1-ax}} \frac{\mathrm{d}v}{1+v^2} \\ &= \int_0^{\frac{x+a}{1-ax}} \frac{\mathrm{d}v}{1+v^2} - \int_0^a \frac{\mathrm{d}v}{1+v^2} = T\left(\frac{x+a}{1-ax}\right) - T(a). \end{aligned}$$

Both arguments of T here are positive, so everything is defined.

The first deduction. Write $X = \frac{x+a}{1-ax}$. By part (i) applied to x and then to a,

$$T(x^{-1}) = T_\infty - T(x) = T_\infty - T(X) + T(a) = T_\infty - T(X) + T_\infty - T(a^{-1}),$$

which is $T(x^{-1}) = 2T_\infty - T\left(\frac{x+a}{1-ax}\right) - T(a^{-1})$.

The second deduction. Given $b > 0$ and $y > \frac{1}{b}$, put $x = \frac{1}{y}$ and $a = \frac{1}{b}$. Then $x > 0$, $a > 0$, and $y > \frac{1}{b}$ is the same as $x < b = \frac{1}{a}$, so the previous result applies. Multiplying top and bottom by by,

$$\frac{x+a}{1-ax} = \frac{\frac{1}{y} + \frac{1}{b}}{1 - \frac{1}{by}} = \frac{b+y}{by-1},$$

and $x^{-1} = y$, $a^{-1} = b$, so

$$T(y) = 2T_\infty - T\left(\frac{y+b}{by-1}\right) - T(b).$$

Part (iii)

$T(\sqrt{3})$**.** Take $y = b = \sqrt{3}$ in the last result; the condition $y > \frac{1}{b}$ holds since $\sqrt{3} > \frac{1}{\sqrt{3}}$. Here

$$\frac{y+b}{by-1} = \frac{2\sqrt{3}}{3-1} = \sqrt{3},$$

so $T(\sqrt{3}) = 2T_\infty - T(\sqrt{3}) - T(\sqrt{3})$, that is $3T(\sqrt{3}) = 2T_\infty$ and

$$T(\sqrt{3}) = \tfrac{2}{3}T_\infty.$$

$T(1)$ **first.** Part (i) with $x = 1$ gives $T(1) = T_\infty - T(1)$, so $T(1) = \frac{1}{2}T_\infty$.

Ask yourself Which value of a makes $\frac{x+a}{1-ax}$ equal to 1 when $x = a$?

We need $\frac{2a}{1-a^2} = 1$, i.e. $a^2 + 2a - 1 = 0$, whose positive root is $a = \sqrt{2} - 1$.

$T(\sqrt{2}-1)$**.** So take $x = a = \sqrt{2} - 1 > 0$ in the addition formula. The condition $x < \frac{1}{a}$ is $\sqrt{2} - 1 < \sqrt{2} + 1$, which holds. Since $a^2 = 3 - 2\sqrt{2}$,

$$\frac{x+a}{1-ax} = \frac{2(\sqrt{2}-1)}{1-(3-2\sqrt{2})} = \frac{2(\sqrt{2}-1)}{2(\sqrt{2}-1)} = 1,$$

so $T(\sqrt{2}-1) = T(1) - T(\sqrt{2}-1)$, giving

$$T(\sqrt{2}-1) = \tfrac{1}{2}T(1) = \tfrac{1}{4}T_\infty.$$

Checks

Of course T is really arctan and $T_\infty = \frac{\pi}{2}$, which we weren't allowed to use but can certainly check against: $\frac{\pi}{3} = \frac{2}{3} \cdot \frac{\pi}{2}$ and $\frac{\pi}{8} = \frac{1}{4} \cdot \frac{\pi}{2}$. Evaluating the integrals numerically (Simpson's rule) gives $T_\infty \approx 1.570796$, $T(\sqrt{3}) \approx 1.047198$ and $T(\sqrt{2}-1) \approx 0.392699$, matching $\frac{2}{3}T_\infty$ and $\frac{1}{4}T_\infty$. The addition formula of part (ii) is the familiar $\tan(A+B)$ formula in disguise, with $x = \tan A$ and $a = \tan B$.

Common slips

- Forgetting to reverse the limits after $u = 1/t$: the minus sign from $\mathrm{d}u = -t^{-2}\,\mathrm{d}t$ is exactly what flips $\int_{\infty}^{1/x}$ to $\int_{1/x}^{\infty}$.
- Quietly using $\tan\frac{\pi}{3} = \sqrt{3}$ or arctan facts in part (iii): the question forbids it, and the point is to get everything from parts (i) and (ii).
- Not checking the conditions $x < \frac{1}{a}$ and $y > \frac{1}{b}$ for the particular numbers used.

Beyond STEP

Defining a function by an integral and deriving all its properties from that definition is how analysis courses build ln, exp, arctan and even π rigorously: here π could simply be defined as $2T_{\infty}$. The substitution $v = \frac{u+a}{1-au}$ is a Möbius transformation, and the fact that it preserves $\frac{\mathrm{d}u}{1+u^2}$ reflects the rotations of a circle. For the analysis side, see Abbott, *Understanding Analysis*; for the geometry of Möbius maps, Needham, *Visual Complex Analysis*.

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